

CHAPTER 7.8

IMPROPER INTEGRALS

Introduction

In previous chapters, we have focused definite integrals with continuous integrands and finite intervals of integration. In this chapter, we will extend the concept of a definite integral to include infinite intervals of integration and integrands that become infinite within the interval of integration.

Improper Integrals

Our main objective in this section is to extend the concept of a definite integral to allow for infinite intervals of integration and integrands with vertical asymptotes within the interval of integration. We will call the vertical asymptotes infinite discontinuities, and we will call integrals with infinite intervals of integration or infinite discontinuities within the interval of integration improper integrals.

Here are some examples:

(1) Improper integrals with infinite intervals of integration:

$$\int_1^{+\infty} \frac{1}{x^2} dx, \quad \int_{-\infty}^0 e^x dx, \quad \int_{-\infty}^{+\infty} \frac{1}{1+x^2} dx$$

(2) Improper integrals with infinite discontinuities in the interval of integration:

$$\int_{-3}^3 \frac{1}{x^2} dx, \quad \int_1^2 \frac{1}{x-1} dx, \quad \int_0^{\pi} \tan x dx$$

(3) Improper integrals with infinite discontinuities and infinite intervals of integration:

$$\int_0^{+\infty} \frac{1}{\sqrt{x}} dx, \quad \int_{-\infty}^{+\infty} \frac{1}{x^2-9} dx, \quad \int_1^{+\infty} \sec x dx$$

Integrals over infinite intervals

7.8.1 Definition

The improper integral of f over the interval $[a, +\infty)$ is defined as

$$\int_a^{+\infty} f(x) dx = \lim_{b \rightarrow +\infty} \int_a^b f(x) dx$$

There are two cases:

- (i) The improper integral is said to **converge** if limit exists and the limit is defined to be the value of the integral.
- (ii) The improper integral is said to **diverge** if limit does not exist. In this case, the integral does not assign a value.

Example 1

Evaluate (a) $\int_1^{+\infty} \frac{1}{x^3} dx$, (b) $\int_1^{+\infty} \frac{1}{x} dx$

Example 2 For what values of p does the integral $\int_1^{+\infty} \frac{1}{x^p} dx$ converge?

7.8.2 Theorem

$$\int_1^{+\infty} \frac{1}{x^p} dx = \begin{cases} \frac{1}{p-1}, & p > 1 \\ \text{diverges}, & p \leq 1 \end{cases}$$

Example 3

Evaluate $\int_0^{+\infty} (1-x)e^{-x} dx$.

7.8.3 Definition

(1) The improper integral of f over the interval $(-\infty, b]$ is defined as

$$\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx$$

The integral is said to **converge** if the limit exists and **diverge** if it does not exist.

(2) The improper integral of f over the interval $(-\infty, b]$ is defined as

$$\int_{-\infty}^{+\infty} f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^{+\infty} f(x) dx = \lim_{a \rightarrow -\infty} \int_a^c f(x) dx + \lim_{b \rightarrow +\infty} \int_c^b f(x) dx$$

where c is any real number. The improper integral is said to **converge** if both terms converge and **diverge** if either term diverges.

Example 4 Evaluate $\int_{-\infty}^{+\infty} \frac{1}{1+x^2} dx$.

7.8.4 Definition

If f is continuous on the interval $[a, b]$, except for an infinite discontinuity at b , then the improper integral of f over the interval $[a, b]$ is defined as

$$\int_a^b f(x) dx = \lim_{k \rightarrow b^-} \int_a^k f(x) dx$$

If limit exists, then the improper integral converges and the limit is defined to be the value of the integral.

If limit does not exist, the improper integral diverges and it has no value.

Example 5 Evaluate $\int_0^1 \frac{dx}{\sqrt{1-x}}$.

7.8.5 Definition

If f is continuous on the interval $[a, b]$, except for an infinite discontinuity at a , then the improper integral of f over the interval $[a, b]$ is defined as

$$\int_a^b f(x) dx = \lim_{k \rightarrow a^+} \int_k^b f(x) dx$$

The integral converges if limit exists and diverges if it does not exist.

If f is continuous on the interval $[a, b]$, except for an infinite discontinuity at a point c in (a, b) , then the improper integral of f over the interval $[a, b]$ is defined as

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

where the two integrals on the right side are themselves improper. The improper integral on the left side is said to converge if both terms on the right side converge and diverge if either term on the right side diverges (**Figure 7.8.9**).

Example 6 Evaluate (a) $\int_1^2 \frac{dx}{1-x}$, (b) $\int_1^4 \frac{dx}{(x-2)^{2/3}}$.

Exercise Set 7.8

Evaluate the following improper integrals:

4. $\int_{-1}^{+\infty} \frac{x}{1+x^2} dx$

We can write
$$\int_{-1}^{+\infty} \frac{x}{1+x^2} dx = \lim_{b \rightarrow +\infty} \int_{-1}^b \frac{x}{1+x^2} dx = \lim_{b \rightarrow +\infty} \left[\frac{1}{2} \ln(1+x^2) \right]_{-1}^b$$
$$= \lim_{b \rightarrow +\infty} \left[\frac{1}{2} \ln(1+b^2) - \ln 2 \right] = \infty$$

Because the integral diverges, it has no value.

6. $\int_0^{+\infty} x e^{-x^2} dx$

We can write
$$\int_0^{+\infty} x e^{-x^2} dx = \lim_{b \rightarrow +\infty} \int_0^b x e^{-x^2} dx = \lim_{b \rightarrow +\infty} \left[-\frac{1}{2} e^{-x^2} \right]_0^b$$
$$= \lim_{b \rightarrow +\infty} \left[-\frac{1}{2} e^{-b^2} + \frac{1}{2} \right]_0^b = \frac{1}{2}$$

Since the limit is finite, the integral converges and its value is $\frac{1}{2}$.

8. $\int_2^{+\infty} \frac{1}{x\sqrt{\ln x}} dx$

We can write
$$\int_2^{+\infty} \frac{1}{x\sqrt{\ln x}} dx = \lim_{b \rightarrow +\infty} \int_2^b \frac{1}{x\sqrt{\ln x}} dx = \lim_{b \rightarrow +\infty} \left[2\sqrt{\ln x} \right]_2^b = 2 \lim_{b \rightarrow +\infty} \left[\sqrt{\ln b} - \sqrt{\ln 2} \right] = \infty$$

Because the integral diverges, it has no value.

10. $\int_{-\infty}^3 \frac{1}{x^2 + 9} dx$

We can write $\int_{-\infty}^3 \frac{1}{x^2 + 9} dx = \lim_{a \rightarrow -\infty} \int_a^3 \frac{1}{x^2 + 9} dx = \lim_{a \rightarrow -\infty} \left[\frac{1}{3} \tan^{-1} \frac{x}{3} \right]_a^3$

$$= \frac{1}{3} \lim_{a \rightarrow -\infty} \left[\frac{\pi}{4} - \tan^{-1} \frac{a}{3} \right] = \frac{\pi}{12} - \frac{1}{3} \left(-\frac{\pi}{2} \right) = \frac{\pi}{4}$$

Since the limit is finite, the integral converges and its value is $\frac{\pi}{4}$.

12. $\int_{-\infty}^0 \frac{e^x}{3 - 2e^x} dx$

We can write $\int_{-\infty}^0 \frac{e^x}{3 - 2e^x} dx = \lim_{a \rightarrow -\infty} \int_a^0 \frac{e^x}{3 - 2e^x} dx = \lim_{a \rightarrow -\infty} \left[-\frac{1}{2} \ln |3 - 2e^x| \right]_a^0$

$$= -\frac{1}{2} [0 - \ln 3] = \frac{1}{2} \ln 3$$

Since the limit is finite, the integral converges and its value is $\frac{1}{2} \ln 3$.

18. $\int_0^8 \frac{dx}{\sqrt[3]{x}}$

We can write $\int_0^8 \frac{dx}{\sqrt[3]{x}} = \lim_{a \rightarrow 0^+} \int_a^8 \frac{dx}{\sqrt[3]{x}} = \lim_{a \rightarrow 0^+} \left[\frac{3}{2} x^{2/3} \right]_a^8$

$$= \frac{3}{2} \lim_{a \rightarrow 0^+} (4 - a^{2/3}) = 6$$

Since the limit is finite, the integral converges and its value is 6.

20. $\int_0^4 \frac{dx}{\sqrt{4-x}}$

We can write $\int_0^4 \frac{dx}{\sqrt{4-x}} = \lim_{k \rightarrow 4^-} \int_0^k \frac{dx}{\sqrt{4-x}} = \lim_{k \rightarrow 4^-} \left[-2\sqrt{4-x} \right]_0^k$

$$= -2 \lim_{k \rightarrow 4^-} (\sqrt{4-k} - 2) = 4$$

Since the limit is finite, the integral converges and its value is 4.

22. $\int_{-3}^1 \frac{x}{\sqrt{9-x^2}} dx$

We can write $\int_{-3}^1 \frac{dx}{\sqrt{9-x^2}} = \lim_{k \rightarrow -3^+} \int_{-3}^1 \frac{dx}{\sqrt{9-x^2}} = \lim_{k \rightarrow -3^+} \left[-\sqrt{9-x^2} \right]_k^1$

$$= (-1) \lim_{k \rightarrow -3^+} (\sqrt{8} - \sqrt{9-k^2}) = -\sqrt{8}$$

Since the limit is finite, the integral converges and its value is $-\sqrt{8}$.

26. $\int_{-2}^2 \frac{dx}{x^2}$

We can write $\int_{-2}^2 \frac{dx}{x^2} = \int_{-2}^0 \frac{dx}{x^2} + \int_0^2 \frac{dx}{x^2}$

$$= \lim_{k \rightarrow 0^-} \int_{-2}^k \frac{dx}{x^2} + \lim_{k \rightarrow 0^+} \int_k^2 \frac{dx}{x^2}$$

$$= \lim_{k \rightarrow 0^-} \left[-\frac{1}{x} \right]_{-2}^k + \lim_{k \rightarrow 0^+} \left[-\frac{1}{x} \right]_k^2$$

$$= (-1) \lim_{k \rightarrow 0^-} \left[\frac{1}{k} + 2 \right] + (-1) \lim_{k \rightarrow 0^+} \left[\frac{1}{2} - \frac{1}{k} \right]$$

$$= \infty$$

Since the limit does not exist, the integral diverges and it has no value.

28. $\int_0^1 \frac{dx}{(x-1)^{2/3}}$

We can write $\int_0^1 \frac{dx}{(x-1)^{2/3}} = \lim_{k \rightarrow 1^-} \int_0^k \frac{dx}{(x-1)^{2/3}}$

$$= \lim_{k \rightarrow 1^-} \left[3(x-1)^{1/3} \right]_0^k$$

$$= 3 \lim_{k \rightarrow 1^-} \left[(k-1)^{1/3} - (-1)^{1/3} \right]$$

$$= 3$$

Since the limit is finite, the integral converges and its value is 3.

29. $\int_0^{+\infty} \frac{dx}{x^2}$

We can write $\int_0^{+\infty} \frac{dx}{x^2} = \int_0^1 \frac{dx}{x^2} + \int_1^{+\infty} \frac{dx}{x^2}$

$$= \lim_{a \rightarrow 0^+} \int_a^1 \frac{dx}{x^2} + \lim_{b \rightarrow +\infty} \int_1^b \frac{dx}{x^2}$$

$$= \lim_{a \rightarrow 0^+} \left[-\frac{1}{x} \right]_a^1 + \lim_{b \rightarrow +\infty} \left[-\frac{1}{x} \right]_1^b$$

$$= \lim_{a \rightarrow 0^+} \left[-1 + \frac{1}{a} \right] + \lim_{b \rightarrow +\infty} \left[-\frac{1}{b} + 1 \right]$$

$$= \infty$$

Since the limit does not exist, the integral diverges and it has no value.

30. $\int_1^{+\infty} \frac{dx}{x\sqrt{x^2-1}}$

We can write $\int_1^{+\infty} \frac{dx}{x\sqrt{x^2-1}} = \int_1^2 \frac{dx}{x\sqrt{x^2-1}} + \int_2^{+\infty} \frac{dx}{x\sqrt{x^2-1}}$

$$= \lim_{a \rightarrow 1^+} \int_a^2 \frac{dx}{x\sqrt{x^2-1}} + \lim_{b \rightarrow +\infty} \int_2^b \frac{dx}{x\sqrt{x^2-1}}$$

$$\begin{aligned}
&= \lim_{a \rightarrow 1^+} [\sec^{-1} x]_a^2 + \lim_{b \rightarrow +\infty} [\sec^{-1} x]_2^b \\
&= \frac{\pi}{3} + \frac{\pi}{2} - \frac{\pi}{3} \\
&= \frac{\pi}{2}
\end{aligned}$$

Since the limit is finite, the integral converges and its value is $\frac{\pi}{2}$.

Note: Many other problems from this Chapter have been solved in the classroom.